

Taha Merghani

taha.y.merghani@gmail.com | taha-y-merghani.github.io | linkedin.com/in/tahamerghani | github.com/taha-y-merghani

Computational linguist with NeurIPS and NAACL publications. Research on dialectal variation using kernel methods translates to understanding how LLMs fail across linguistic contexts. Built Siri debugging tools still in production eight years later. Seeking research or software engineering roles in LLM evaluation and robustness. Open to interesting opportunities elsewhere.

Experience

Software Engineering Intern, *Apple*

Cupertino, CA

Summer 2016

- Built internal debugging and visualization tools for Siri NLU pipeline; tools remain in production use eight years later

Research Intern, *MIT CSAIL*

Cambridge, MA | Advisor: James Glass

Summer 2015

- Developed lexical models for Egyptian Arabic ASR; demonstrated grapheme lexicon outperformed diacritized approaches

Graduate Research Assistant, *Georgia Institute of Technology*

Atlanta, GA

Aug 2017 – May 2019

- Published at NeurIPS 2018 Workshop (geolinguistic variation via kernel methods) and NAACL 2018 Workshop (stylistic variation in POS tagging)
- Research on human-computer interaction for conversational AI with Dr. Mark Riedl

AI Software Engineer, *Decooda*

Atlanta, GA

Jul 2019 – Aug 2020

- Built ETL pipelines using Matillion and Snowflake for customer analytics and ML model training

Solutions Architect, *Mesa Associates*

Remote

Nov 2022 – Apr 2023

- Deployed computer vision models on Oracle Cloud for automated electrical equipment inspection; built Python tooling for drone image processing

Education

M.S. Computer Science *Georgia Institute of Technology*

GPA: 3.87/4.0 | Advisors: Jacob Eisenstein, Mark Riedl

Aug 2017 – May 2019

B.S. Computer Engineering *Jackson State University*

GPA: 4.0/4.0 | Jacksonian Award: Highest GPA in College of Science, Engineering, and Technology

Aug 2013 – May 2017

Publications

- Balusu, Merghani, Eisenstein. Stylistic Variation in Social Media Part-of-Speech Tagging. [NAACL Workshop 2018](#).
- Merghani, Eisenstein. Geolinguistic Analysis via Twitter. [NeurIPS Workshop \(Black in AI\) 2018](#).

Selected Writing

- [What 5,000 Hours of Mastering Tekken Taught Me About AI Research](#), #1 on r/ArtificialIntelligence on a day
- [From Sudan to Silicon Valley: Beyond the Resume](#). Jeff Dean shared (200k on twitter)

Skills

Technical: Python, TypeScript, SQL, PyTorch, Hugging Face, scikit-learn, spaCy, kernel methods, Snowflake, AWS, Docker

Languages: Arabic (native), English (near-native)